EXACTNESS OF 0 -> PIC(T) -> PIC(X_T) -> PICARD FUNCTOR UNDER O_T = F_*O_X_T

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Allowed standard prerequisites (editorial).

- Situation situation-pic (canonical tag 0D25). cited at statement line 2979. Standing situation: f : X -> B with the structure sheaf pushforward an isomorphism.
- Proposition descent-proposition-fpqc-descent-quasi-coherent (canonical tag 023T). cited at proof line 3018. fpqc/fppf descent of quasi-coherent modules: descent data are effective and the descent functor is fully faithful.

Source reference. Original definition 0D25: sources/stacks-chow/tags.txt, source lines 16370–16370. Cited at statement line 2979: the lemma is phrased in this situation; load-bearing.

Source reference. Original proposition 023T: sources/stacks-chow/tags.txt, source lines 2754–2754. Cited at proof line 3018: the cocycle f_ij defines an invertible O_B-module N by descent, and the functor is fully faithful giving f^*N = L; load-bearing.

Lemma 0.2. *In Situation 0D25. If $\mathcal{O}_T \to f_{T,*}\mathcal{O}_{X_T}$ is an isomorphism for all schemes T over B, then*

$$0 \to \mathrm{Pic}(T) \to \mathrm{Pic}(X_T) \to \mathrm{Pic}_{X/B}(T)$$

is an exact sequence for all T.

Proof. We may replace B by T and X by X_T and assume that $B = T$ to simplify the notation. Let $\mathcal{N}$ be an invertible $\mathcal{O}_B$-module. If $f^*\mathcal{N} \cong \mathcal{O}_X$, then we see that $f_*f^*\mathcal{N} \cong f_*\mathcal{O}_X \cong \mathcal{O}_B$ by assumption. Since $\mathcal{N}$ is locally trivial, we see that the canonical map $\mathcal{N} \to f_*f^*\mathcal{N}$ is locally an isomorphism (because $\mathcal{O}_B \to f_*f^*\mathcal{O}_B$ is an isomorphism by assumption). Hence we conclude that $\mathcal{N} \to f_*f^*\mathcal{N} \to \mathcal{O}_B$ is an isomorphism and we see that $\mathcal{N}$ is trivial. This proves the first arrow is injective.

Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module which is in the kernel of $\mathrm{Pic}(X) \to \mathrm{Pic}_{X/B}(B)$. Then there exists an fppf covering $\{B_i \to B\}$ such that $\mathcal{L}$ pulls back to the trivial invertible sheaf on X_{B_i}. Choose a trivializing section s_i. Then $\mathrm{pr}_0^*s_i$ and $\mathrm{pr}_1^*s_j$ are

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both trivialising sections of $\mathcal{L}$ over $X_{B_i \times_B B_j}$ and hence differ by a multiplicative unit

$$f_{ij} \in \Gamma(X_{S_i \times_B B_j}, \mathcal{O}^*_{X_{B_i \times_B B_j}}) = \Gamma(B_i \times_B B_j, \mathcal{O}^*_{B_i \times_N B_j})$$

(equality by our assumption on pushforward of structure sheaves). Of course these elements satisfy the cocycle condition on $B_i \times_B B_j \times_B B_k$, hence they define a descent datum on invertible sheaves for the fppf covering $\{B_i \to B\}$. By Descent, Proposition 023T there is an invertible $\mathcal{O}_B$-module $\mathcal{N}$ with trivializations over B_i whose associated descent datum is $\{f_{ij}\}$. (The proposition applies because B is a scheme by the replacement performed at the start of the proof.) Then $f^*\mathcal{N} \cong \mathcal{L}$ as the functor from descent data to modules is fully faithful. $\square$